

Object Pose Estimation from Silhouette using Differentiable Rendering

Madiyar Bolatov, AAI-2501M

March 25, 2026

1 Introduction

The goal of this project is to implement an algorithm for estimating the pose of a 3D object by optimizing the camera position using only a silhouette image. The approach is based on differentiable rendering, which allows gradients to propagate through the rendering pipeline and enables optimization of camera parameters.

Unlike traditional methods that rely on feature matching or RGB information, this method uses only binary silhouette data, making it robust to texture variations.

2 Methodology

2.1 Object Preparation



Figure 1: Object Guitar

A custom 3D model (guitar) in .obj format was used instead of the default teapot from the tutorial. Since the model did not include textures, a simple uniform gray texture was applied using TexturesVertex.

To ensure stable optimization, the mesh was normalized: centered at the origin; scaled to fit within a unit sphere.

Additionally, mesh simplification (**decimation**) was applied to reduce computational complexity.

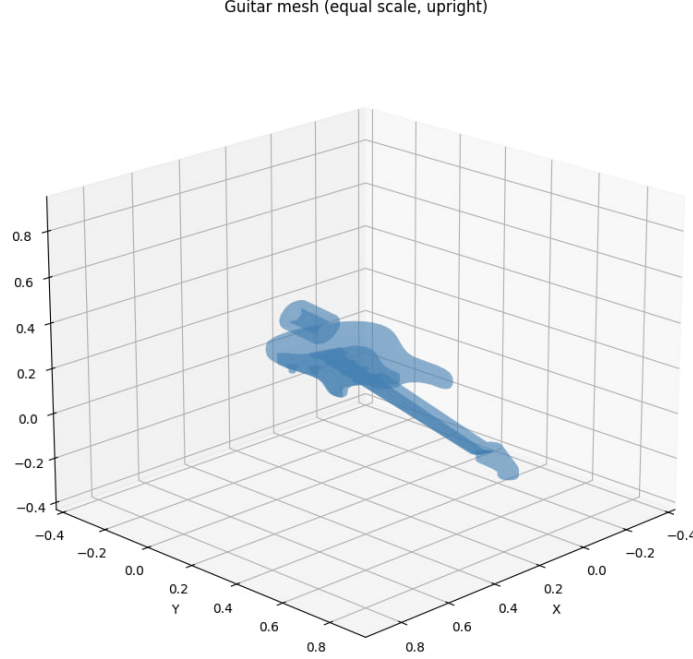


Figure 2: After normalization and decimation

2.2 Rendering Setup

Two renderers were defined:

- Silhouette renderer using **SoftSilhouetteShader**
- RGB renderer using **HardPhongShader** (for visualization only)

Key parameters:

- `image_size = 128`
- $\sigma = 10^{-4}$
- `faces_per_pixel = 10`

The silhouette image is obtained from the alpha channel of the rendered output.

2.3 Camera Model

The camera is parameterized by its position in 3D space: $c = (x, y, z)$

Rotation and translation matrices are computed using a look-at transformation. The camera is always directed toward the object center.

2.4 Optimization Objective

The objective is to minimize the difference between the predicted silhouette and the target silhouette:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \left(S_{\text{pred}}^{(i)} - S_{\text{target}}^{(i)} \right)^2 \quad (1)$$

where:

- S_{pred} — predicted silhouette,
- S_{target} — target silhouette.

The optimization is performed using the Adam optimizer.

3 Experimental Setup

Target camera position: (0.0, 0.0, 2.0)

Initial camera position: (2.5, -1.5, 3.0)

The initial position was intentionally chosen far from the target to evaluate the convergence behavior of the optimization process.

4 Results

4.1 Loss Convergence

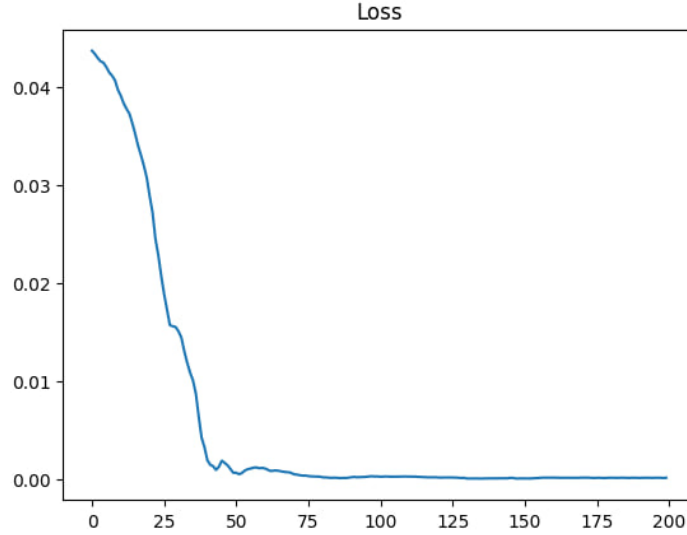


Figure 3: Loss Graph

The loss decreased rapidly from approximately 0.0436 to less than 0.0001, demonstrating stable convergence.

Fast decrease in early iterations (40 iterations) Smooth convergence afterward

4.2 Pose Estimation Accuracy

Target camera: (0.0, 0.0, 2.0)

Optimized camera: (0.0051, 0.0041, 1.9989)

The error is negligible, indicating highly accurate pose recovery.

4.3 Visual Comparison

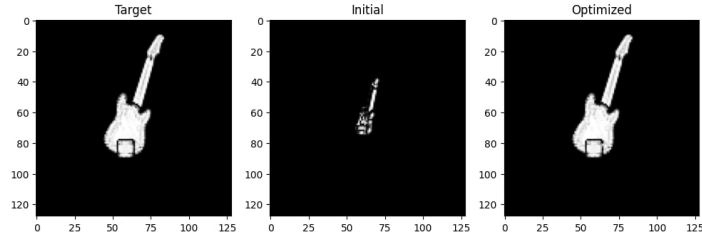


Figure 4: Visual Comparison

1. Initial silhouette: object appears small and misaligned
2. Optimized silhouette: nearly identical to the target
3. RGB visualization confirms correct pose alignment

5 Challenges and Solutions (Individual Part)

5.1 Challenges

1. Mesh Scale Issues
 - Without normalization, the object was either too large or too small.
 - This caused incorrect silhouettes and unstable gradients.
2. Camera Initialization
 - If the camera was too far, the object became barely visible.
 - If too close, parts of the object were clipped.
3. Renderer Parameters
 - Small sigma values produced sharp silhouettes but unstable gradients.
 - Low faces_per_pixel caused poor silhouette quality.

6 Conclusion

In this project, a differentiable rendering approach was successfully implemented to estimate object pose using only silhouette information. The method demonstrated:

- high accuracy in camera position recovery,
- stable and fast convergence,
- robustness to the absence of texture information.

This confirms that silhouette-based pose estimation is effective when combined with differentiable rendering techniques.